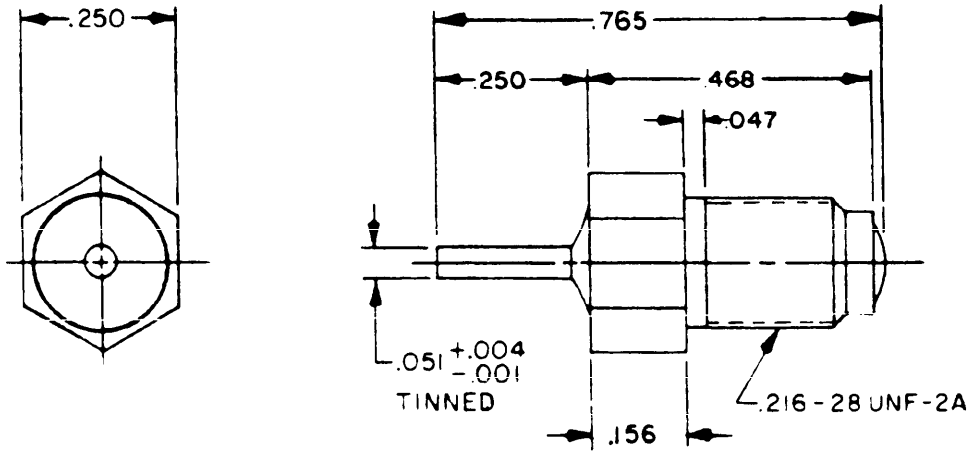
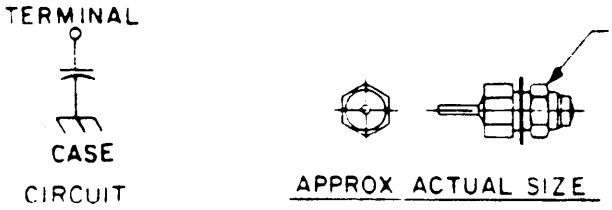
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC, (GENERAL PURPOSE), ESTABLISHED RELIABILITY STYLE CKR82 (STANDOFF)

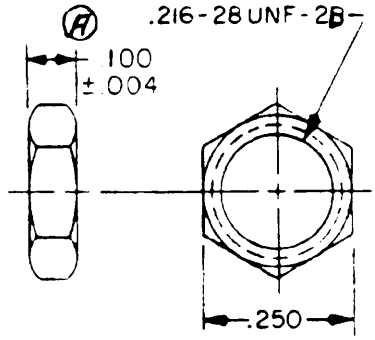
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39014.





INCHES	MM	INCHES	MM
.001	.03	.156	3.96
.004	.10	.216	5.46
.047	1.19	.250	6.35
.051	1.30	.468	11.89
.100	2.54	.765	19.43



DETAIL-B-
MOUNTING NUT

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.031(.79 mm).
3. A nut and lock washer shall be supplied with each capacitor.

FIGURE 1. Style CKR82 capacitors.

(A) denotes changes

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:26Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Standoff.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +85°C.

Voltage rating: 500 volts dc.

Dissipation factor: Not more than 3.0 percent.

Dielectric withstanding voltage:

Application of test voltage: Between one capacitor-element terminal and the case.

Barometric pressure:

Method 105, MIL-STD-202, condition B (50,000 feet).
Application of test voltage: Between one terminal and the case.
Test potential: 150 percent of dc rated voltage.

Insulation resistance:

Measured between one terminal and the case:
At 25°C: Not less than 100,000 megohms.
At 85°C: Not less than 10,000 megohms.

Insertion loss: Not applicable.

Seal: Not applicable.

Solderability: Method 208, MIL-STD-202, one terminal only.

Voltage-temperature limits: In accordance with MIL-C-39014.

Capacitance change with reference to 25°C		
Steps A through D of table V Bias = 0 volts	Steps E through G of table V Bias = rated voltage	For capacitance
±15 percent	+15, -25 percent	Less than 1,000 pF
+22, -56 percent	+22, -66 percent	1,000 pF and greater

Immersion:

IR: Not less than 3,000 megohms.
(A) ΔC: ±20 percent from initial measured value.

Salt spray (corrosion): In accordance with MIL-C-39014.

Terminal strength: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:26Z
Check the source to verify that this is the current version before use.

Moisture resistance:

Application of test voltage: Between one terminal and the case.
IR: Not less than 3,000 megohms.
ΔC: ±20 percent from initial measured value.

Resistance to soldering heat:

ΔC: Less than ±20 percent from initial measured value.
Cooling time prior to measurement after test: 1 ±.1 hour.

Life:

Rated conditions: 100 percent of dc rated voltage at 85°C.

(A) 0 through 3,000 hours:

IR: At 25°C not less than 50,000 megohms.
At 85°C not less than 5,000 megohms.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 3.0 percent.

4,000 through 32,000 hours:

IR: At 25°C not less than 10,000 megohms.
At 85°C not less than 1,000 megohms.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 3.0 percent.

Accelerated conditions, 200 percent of dc rated voltage at 85°C.

0 and 250 hours:

IR: At 25°C not less than 50,000 megohms.
At 85°C not less than 5,000 megohms.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 3.0 percent.

1,000, 2,000 and 4,000 hours:

IR: At 25°C not less than 10,000 megohms.
At 85°C not less than 1,000 megohms.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 3.0 percent.

Marking: Marking shall consist of the "J" certification mark and the military part number as shown on figure 2.

(A)

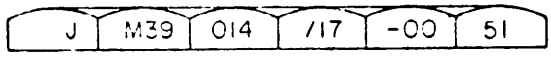


FIGURE 2. Hex flats - expanded view.

Part number: M39014/17- (dash number from table I).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:26Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics and dash number.

(A)

Capacitance	Capacitance tolerance	Dash number			
		Failure rate level (percent/1,000 hours)			
		1.0 (M)	0.1 (P)	0.1 (R)	0.001 (S)
pF	± percent				
10	10	0051	0101	0151	0201
10	20	0052	0102	0152	0202
12	10	0053	0103	0153	0203
15	10	0054	0104	0154	0204
15	20	0055	0105	0155	0205
18	10	0056	0106	0156	0206
22	10	0057	0107	0157	0207
22	20	0058	0108	0158	0208
27	10	0059	0109	0159	0209
33	10	0060	0110	0160	0210
33	20	0061	0111	0161	0211
39	10	0062	0112	0162	0212
47	10	0063	0113	0163	0213
47	20	0064	0114	0164	0214
56	10	0065	0115	0165	0215
68	10	0066	0116	0166	0216
68	20	0067	0117	0167	0217
82	10	0068	0118	0168	0218
100	10	0069	0119	0169	0219
100	20	0070	0120	0170	0220
120	10	0071	0121	0171	0221
150	10	0072	0122	0172	0222
150	20	0073	0123	0173	0223
180	10	0074	0124	0174	0224
220	10	0075	0125	0175	0225
220	20	0076	0126	0176	0226
270	10	0077	0127	0177	0227
330	10	0078	0128	0178	0228
330	20	0079	0129	0179	0229
390	10	0080	0130	0180	0230
470	10	0081	0131	0181	0231
470	20	0082	0132	0182	0232
560	10	0083	0133	0183	0233
680	10	0084	0134	0184	0234
680	20	0085	0135	0185	0235
820	10	0086	0136	0186	0236
1,000	+80, -20	0087	0137	0187	0237
1,200	+80, -20	0088	0138	0188	0238
1,500	+80, -20	0089	0139	0189	0239

Source: https://assist.dla.mil -- Downloaded: 20260-21T17:26Z
Check the source to verify that this is the current version before use.

(A) TABLE II. Substitutability data.

Dash numbers in MIL-C-39014/17A 1/				Substitutable for dash number in MIL-C-39014/17				
M	P	R	S	L	M	P	R	S
0051	0101	0151	0201	0001	0051	0101	0151	0201
0052	0102	0152	0202	0002	0052	0102	0152	0202
0053	0103	0153	0203	0003	0053	0103	0153	0203
0054	0104	0154	0204	0004	0054	0104	0154	0204
0055	0105	0155	0205	0005	0055	0105	0155	0205
0056	0106	0156	0206	0006	0056	0106	0156	0206
0057	0107	0157	0207	0007	0057	0107	0157	0207
0058	0108	0158	0208	0008	0058	0108	0158	0208
0059	0109	0159	0209	0009	0059	0109	0159	0209
0060	0110	0160	0210	0010	0060	0110	0160	0210
0061	0111	0161	0211	0011	0061	0111	0161	0211
0062	0112	0162	0212	0012	0062	0112	0162	0212
0063	0113	0163	0213	0013	0063	0113	0163	0213
0064	0114	0154	0214	0014	0064	0114	0164	0214
0065	0115	0165	0215	0015	0065	0115	0165	0215
0066	0116	0166	0216	0016	0066	0115	0166	0216
0067	0117	0167	0217	0017	0067	0117	0167	0217
0068	0118	0168	0218	0018	0068	0118	0168	0218
0069	0119	0169	0219	0019	0069	0119	0169	0219
0070	0120	0170	0220	0020	0070	0120	0170	0220
0071	0121	0171	0221	0021	0071	0121	0171	0221
0072	0122	0172	0222	0022	0072	0122	0172	0222
0073	0123	0173	0223	0023	0073	0123	0173	0223
0074	0124	0174	0224	0024	0074	0124	0174	0224
0075	0125	0175	0225	0025	0075	0125	0175	0225
0076	0126	0176	0226	0026	0076	0126	0176	0226
0077	0127	0177	0227	0027	0077	0127	0177	0227
0078	0128	0178	0228	0028	0078	0128	0178	0228
0079	0129	0179	0229	0029	0079	0129	0179	0229
0080	0130	0180	0230	0030	0080	0130	0180	0230
0081	0131	0181	0231	0031	0081	0131	0181	0231
0082	0132	0182	0232	0032	0082	0132	0182	0232
0083	0133	0183	0233	0033	0083	0133	0183	0233
0084	0134	0184	0234	0034	0084	0134	0184	0234
0085	0135	0185	0235	0035	0085	0135	0185	0235
0086	0136	0186	0236	0036	0086	0136	0186	0236
0087	0137	0187	0237	0037	0087	0137	0187	0237
0088	0138	0188	0238	0038	0088	0138	0188	0238
0089	0139	0189	0239	0039	0089	0139	0189	0239

1/ Dash numbers in lower FR level columns are substitutable for dash numbers in all high FR level columns.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:26Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 11
NASA - NA

Review activities:
Navy - OS, TD
DLA - ES

User activities:
Navy - AS, SH, MC

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5910-1508-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:26Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-39014/17A	CAPACITORS, FIXED, CERAMIC DIELECTRIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:26Z
Check the source to verify that this is the current version before use.